

ANN masking and PING encoding-rate psychometric curves

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Abstract

A pyramidal–interneuron gamma (PING) network trained on clean grayscale handwritten digits encounters two different forms of weak evidence: fewer Poisson-encoded spikes, or permanent removal of foreground pixels. An architecture-matched artificial neural network (ANN) provides the non-spiking reference. The frozen PING classifier stays at chance through 0.5 Hz and reaches 49.7% correct at 2 Hz. Under matched spatial masking, PING outperforms the ANN at intermediate deletion but reaches chance sooner: at retention $q = 0.02$, PING is at 10.7% while the ANN remains at 21.7%. The calibrated range now bounds a follow-up training experiment with variable evidence.

Methods

The dataset is MNIST, a ten-class collection of grayscale handwritten digits. The standard corpus is split into 56000 training and 14000 held-out examples. The protocol has three ordered parts:

1. **Calibrate foreground deletion with an ANN.** Train 3 seeds of a non-spiking classifier with 784 inputs, one rectified-linear-unit (ReLU) hidden layer of 1024 units (matching the PING excitatory population), and 10 outputs. Training uses 15 epochs, batches of 256, and learning rate 0.001. Binarize each held-out image at intensity 0, then retain every foreground pixel independently with probability q . Here $q = 1$ leaves the foreground intact and $q = 0$ removes it. Evaluate 10 mask draws per image at every registered q .
2. **Measure the frozen PING encoding-rate curve.** Hold presentation at 200 ms and sweep the maximum per-pixel Poisson encoding rate. Newly simulate the low-rate cells and reuse the matching published cells from exp048 at higher rates. No PING weight changes during this experiment.
3. **Compare spatial masking directly.** Select 100 fixed held-out images. Present the identical binarized images and Bernoulli masks to every ANN and PING seed. PING uses a maximum encoding rate of 25 Hz and its trained presentation duration. This paired subset makes differences attributable to the classifiers rather than different corruptions.

Results

The ANN needs almost complete foreground removal to reach chance

The architecture-matched ANN remains above chance until foreground retention falls to $q = 0.005$, fewer than one visible foreground pixel per image on average.

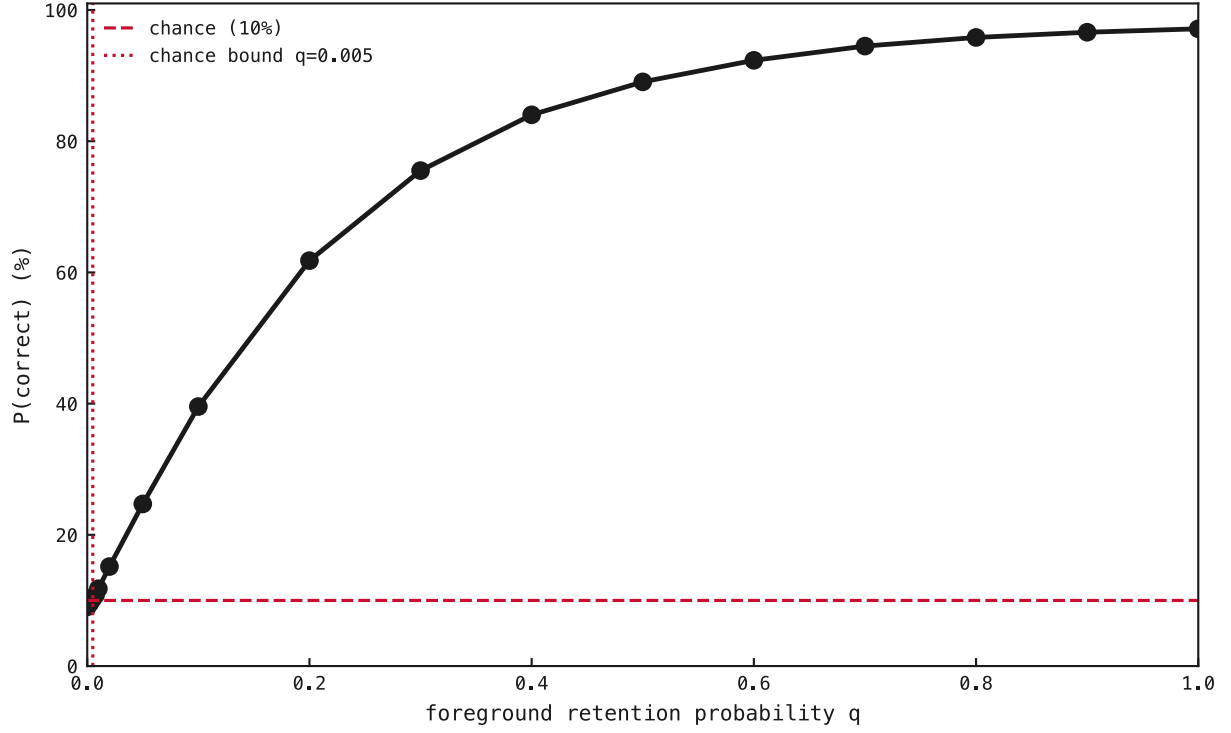


Figure 1: Held-out ANN classification accuracy as foreground evidence is removed. The horizontal axis is foreground-retention probability q (dimensionless); the vertical axis is the probability of a correct classification, in percent. Points are means across the architecture-matched ANN seeds and the band is one standard error across seeds. The dashed red line is ten-class chance performance; the dotted line marks the chance-region bound read from the run. Accuracy stays above chance until almost every foreground pixel is absent.

PING has an empty-input floor followed by a steep rate transition

The frozen network remains on its zero-input floor through 0.5 Hz. Its accuracy then rises from 16.7% at 1 Hz to 67.3% at 3 Hz, before joining exp048’s 79.1% point at 5 Hz.

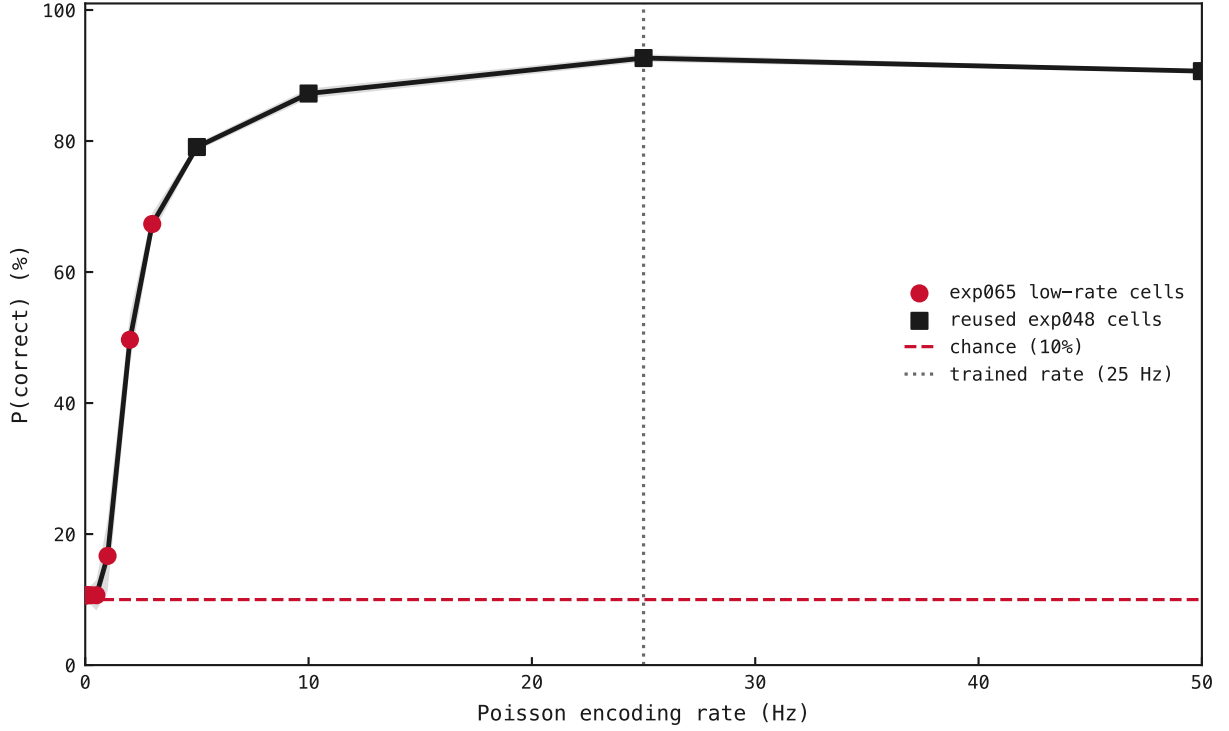


Figure 2: Held-out classification accuracy of the existing frozen PING networks against input strength. The horizontal axis is the maximum per-pixel Poisson encoding rate in hertz; the vertical axis is the probability of a correct classification, in percent. Presentation lasts 200 ms. Red circles are newly simulated low-rate cells; black squares reuse published exp048 cells, with marker shape preserving the source distinction without colour. Points are means across the trained seeds and the band is one standard error. The dashed line marks ten-class chance and the dotted line the 25 Hz training rate. Below the onset the encoder usually supplies no spikes; classification then rises steeply between 1 and 5 Hz.

PING is competitive at intermediate masking but reaches chance sooner

The matched comparison does not have one uniformly better classifier. With 29.5 visible pixels on average ($q = 0.2$), PING reaches 68.7% against the ANN’s 60.7%. They coincide near 45% at $q = 0.1$. PING still leads at $q = 0.05$, then reaches chance at $q = 0.02$ while the ANN remains above it. The PING transition is steeper, not uniformly worse.

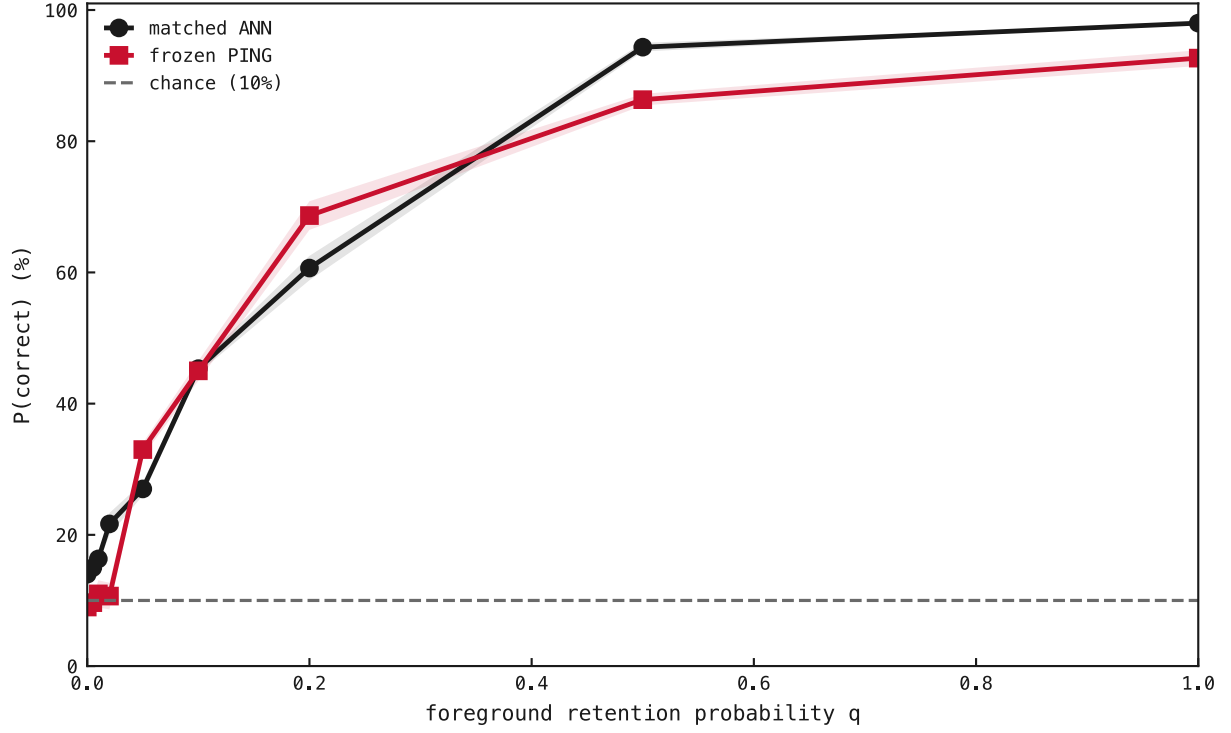


Figure 3: Architecture-matched ANN and frozen PING classification under identical spatial deletion. The horizontal axis is foreground-retention probability q (dimensionless); the vertical axis is the probability of a correct classification, in percent. Black circles denote ANN and red squares PING, so marker shape also identifies each classifier in grayscale. Each point uses 100 fixed held-out examples and the same Bernoulli foreground masks; uncertainty is one standard error across trained seeds. PING runs at 25 Hz for 200 ms. PING is competitive at intermediate retention but crosses into its chance region before the ANN.

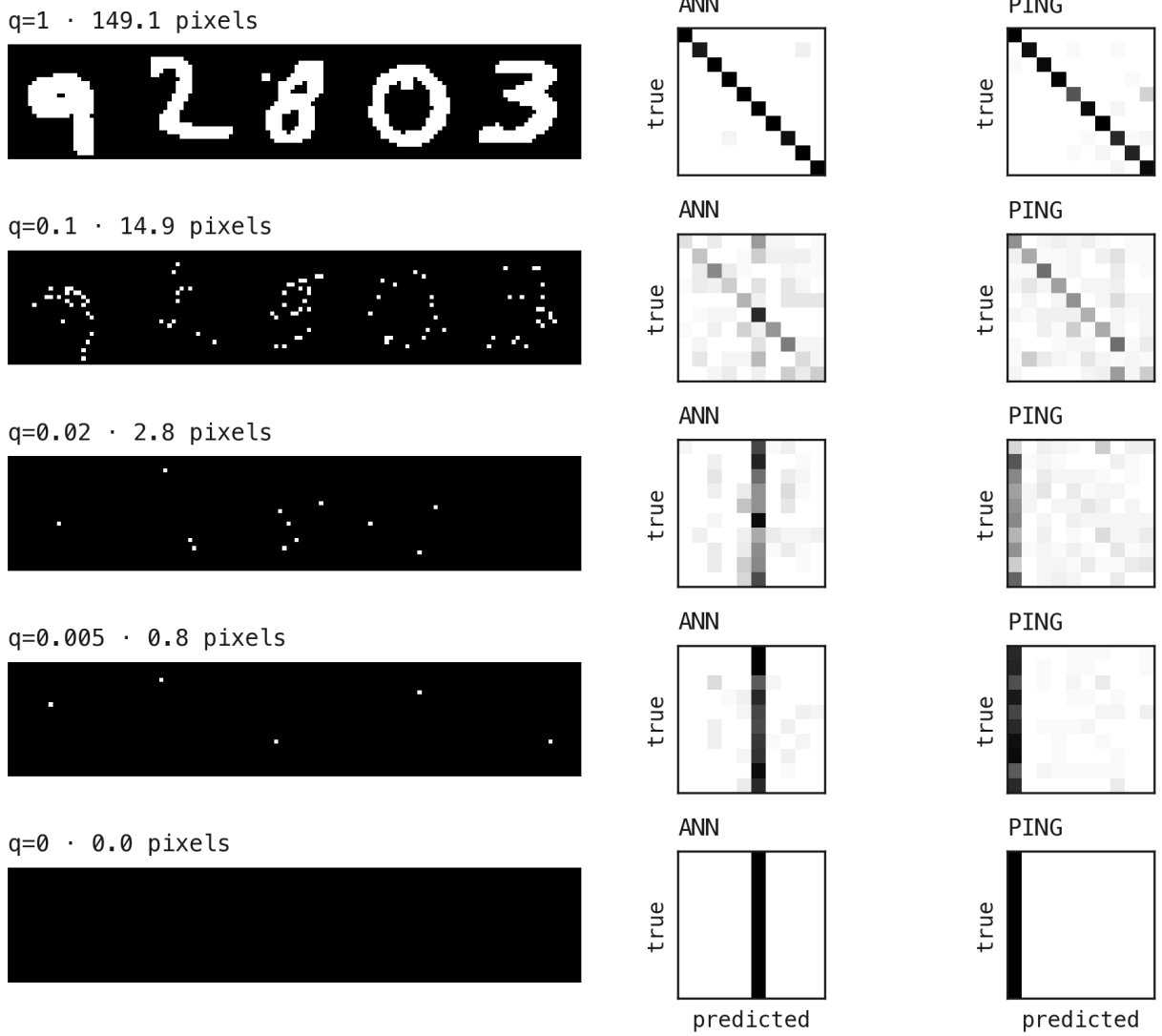


Figure 4: Stimulus and error structure across the matched masking curve. Rows progress from intact input through intermediate masking, the PING and ANN chance regions, and the blank control; the left panel in each row shows five example digits and reports q plus the mean number of visible foreground pixels. The middle and right panels are row-normalized ANN and PING confusion matrices: rows are true classes and columns predicted classes. The blank-input columns reflect each model’s constant prediction and the class prevalence in this fixed subset, not evidence extracted from a blank image.

Caveat. Binarization maps every nonzero antialiased MNIST pixel to full intensity. At $q = 1$ that drives PING more strongly than the original grayscale image at the same encoding-rate ceiling. This curve therefore measures robustness to permanent spatial deletion from a binarized stimulus; it is not a second measurement of grayscale contrast.

The chance bounds define the training range and explain exp048’s failures

The ANN reaches chance near $q = 0.005$; matched PING by $q = 0.02$. These bounds set the range for variable-evidence training. Encoding-rate reduction and spatial masking remain distinct because they remove different evidence. For rate sweeps, 2 Hz is the lowest tested point with clearly useful population-level information, while 5 Hz is a more practical lower

operating point; denser sampling below 0.5 Hz would mostly characterize empty encoder output.

This also changes the interpretation of exp048's two errors in its variable-condition stream. The failed 5 was presented for 200 ms at 10 Hz, a condition at which the population accuracy here is 87.3%, far above chance. The failed 7 at 15 Hz and 75 ms likewise lies above the empty-input rate floor, although its shorter integration window supplies less total evidence. Those examples are therefore natural trial-level classification failures under weak evidence, not evidence that their encoding rates are generally too low to represent the digits. Encoding rate should only be discarded as uninformative below the population-level transition measured here, rather than whenever an individual example is misclassified.